

Interaction-Conditioned Residual Composition for Cooperative Multi-USV Target Defense

Manuscript in preparation

Abstract—Cooperative defense requires a pursuit maneuver to induce an appropriate response from the remaining vessels. We present interaction-conditioned residual composition for multi-USV target defense. A joint interception assignment provides the control prior. Selecting pursuit residuals conditionally reallocates the remaining guard roles, and an adaptive composition selects both the pursuing set and the teammate response through sustained closed-loop prediction. A matched additive control retains the same candidates and forecast budget, isolating the contribution of joint-composition value. In 256 unused six-defender scenarios at attacker thrust multipliers 4 and 6, adaptive joint selection achieves 256 captures, no breaches, and no collisions, with a mean capped capture time of 21.103 s. It reduces time by 20.12% against matched additive selection and by 30.23–31.99% against matched short prediction, fixed initial roles, and an existing long predictor. All four contrasts pass the declared task criteria and an additional analysis that accounts for linked random streams across agility strata. Fixed guard completion remains a faster comparison on this bank; adaptation is assessed through its speed and reliability together. We also retain a residual actor–critic realization with paired conditional-return supervision. The current learning pilots have not established additional task gains, so the reported advantages belong to predictive conditional composition.

Index Terms—Multi-robot systems, unmanned surface vehicles, cooperative control composition, residual control, predictive control, conditional value learning.

I. INTRODUCTION

Multi-USV defense against an agile intruder requires sustained coordination of the team’s motion around a protected region. When a defender leaves an interception role to pursue, its teammates may need to cover the resulting gap. Each local control choice must therefore suit both the vessel’s state and the team’s response. We study cooperative control composition in this setting: how should local pursuit choices and concurrent teammate controls be coordinated to produce effective team defense? Composition variables parameterize the choice; the research object is how the team’s control sources are used together. Predictive selection and learned blending provide two realizations, with separate evidence for their task outcomes.

Marine autonomy connects perception, decision making, and control [1]. Territory guarding, distributed pursuit, reach-avoid games, and faster-evader pursuit provide complementary formulations of coordinated defense [2], [3], [4], [5]; vessel dynamics and shared space couple the consequences of individual maneuvers [6], [7]. Reinforcement learning supports decentralized pursuit [8], aerial pursuit–evasion [9], multi-USV hunting [10], and integrated marine decision making and control [11]. Shared policies and mean embeddings accommodate variable-size teams [12], while joint values [13]

and communication representations [14], [15] support learning from team interactions. These tools motivate representing the particular interaction relevant to composition: the relation between the control candidates available to different defenders at the same instant.

Bayesian controller fusion [16], residual model predictive control [17], and ARBoids [18] combine structured control with learned adaptation. ARBoids already achieves cooperative defense through state-conditioned individual blending, teammate observations, the Boids prior, a shared policy, and task feedback. It learns when each defender should rely more on learned or rule-based control. Our learning realization retains its control sources and 512-unit backbone, and investigates how local blending decisions should depend on concurrent candidate controls and their conditional long-term team consequences. The predictive realization instead makes the induced teammate response explicit in a structured residual operator. The distinction is the explicit policy representation and value learning of composition interactions within an already cooperative system.

PSAC-MATD3 [19] addresses multi-USV cooperative policy learning through policy sharing, phase-aware rewards, prioritized replay, and CBF correction of execution commands. Our mechanism acts on the composition of heterogeneous control sources and the value of changing that composition. Both approaches concern cooperative control; their methodological emphasis lies at different points in the learning and control process.

Two interactions must be distinguished. Candidate-control interaction is represented in the policy input and conditions the blending decision. Configuration interaction concerns whether the value of one defender’s composition change depends on its teammates’ composition choices. Team rewards evaluate the joint outcome, but do not directly isolate this conditional difference. Rollout evaluation [20], safe learning [21], and trajectory prediction for USV cooperation [22] motivate learning from future consequences. Counterfactual multi-agent learning [24] motivates holding teammates’ current choices fixed, and model-based value expansion [25] motivates truncated rollouts with a learned tail. We apply these ideas specifically to the blending variable: candidate controls remain fixed at the intervention step, and all defenders subsequently resume closed-loop decisions in each branch.

The central contribution is a conditional residual operator: choosing which defenders pursue changes how the remaining defenders guard, and joint value evaluates the resulting team response. An adaptive version selects the response mode together with the pursuit composition. Matched joint,

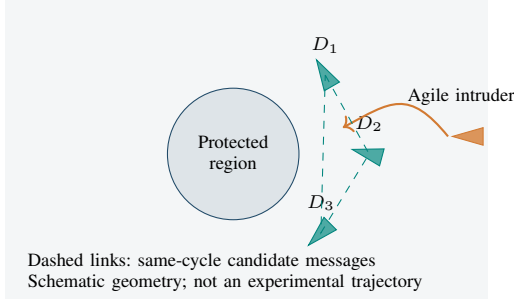


Fig. 1. Cooperative control composition for target defense. Dashed links carry concurrent candidate-control information; the drawing assigns no fixed maneuver roles. Defense succeeds through capture or survival to the 60-s deadline, with region entry and defender collision recorded separately.

additive, and short-horizon controls test whether sustained interaction evaluation improves the task while retaining the same candidate controls. We also investigate a trainable realization that represents concurrent candidate relations in a two-stage residual actor and supervises joint critics with paired conditional-return differences. These are distinct realizations of cooperative composition, with separate empirical claims. The completed predictive tests establish task gains; the current learning pilots do not establish that those gains have been acquired by the residual actor-critic.

II. TASK AND CONTROL MODEL

A. Dynamics, observations, and outcomes

Let N homogeneous defenders protect a circular target region of radius 15 m against one intruder. The numerical environment follows the released three-degree-of-freedom WAM-V model [18],

$$\dot{\eta}_i = R(\psi_i)\nu_i, \quad M\dot{\nu}_i + C(\nu_i)\nu_i + D(\nu_i)\nu_i = \tau_i, \quad (1)$$

with the source current perturbations and a 0.05-s integration step. A command is held for four integration steps, giving a 0.2-s control period. Differential thrust lies in $[-500, 1000]$ N per thruster; intruder agility scales its thrust bounds. Observations contain target-relative and intruder-relative geometry, measured intruder velocity, Boids features, and teammate distance-bearing pairs. Candidate messages add measured defender position, heading, translational velocity, yaw rate, and both current action candidates. No future disturbance, true intruder agility, or centralized training state is an actor input.

We use `paper-parameters-v1`: a run terminates at actual target entry ($d_{AT} \leq 15$ m), a defender separation at or below 5 m, capture ($\min_i d_{iA} < 5$ m), or 60 s. This is also the priority order for simultaneous terminal events. Independent physical collision and breach flags are retained alongside that exclusive outcome. The deadline is an absorbing task termination, with zero bootstrap value.

B. Control sources and nominal composition

For the learning realization, Boids supplies pursuit, separation, alignment, and cohesion forces using the released parameters. A shared learned candidate $\ell_i \in (-1, 1)^2$ maps to

physical thrust $u_i^L = 750\ell_i + 250$. The mixed nominal control is

$$u_i^{\text{nom}} = \theta_i u_i^L + (1 - \theta_i) u_i^B, \quad 0 \leq \theta_i \leq 1. \quad (2)$$

We define *control composition* as the state-dependent combination of control sources in the nominal command. The predictive realization in Section III uses an interception prior and pursuit feedback; the learning realization uses the sources in (2). The candidate controls and team composition configuration are

$$\mathcal{A} = ((u_i^L, u_i^B))_{i=1}^N, \quad \Theta = (\theta_1, \dots, \theta_N). \quad (3)$$

Cooperative control composition seeks to match these local choices to inter-vessel interactions and the team task. Each blending coefficient describes the relative contribution of two sources to one nominal command: $\theta_i + (1 - \theta_i) = 1$. There is no shared cross-vessel coefficient budget or constraint $\sum_i \theta_i = 1$. Local control authority is an interpretation of this relative contribution, not a transfer of actuator permissions.

For fixed candidate controls,

$$\frac{\partial u_i^{\text{nom}}}{\partial \theta_i} = u_i^L - u_i^B. \quad (4)$$

Thus coefficient variation translates into command variation through candidate disagreement. Every primary learning method then uses the same existing second-order CBF filter, with a 7-m barrier separation, the released relaxed quadratic-program settings, and identical physical saturation. Writing $\mathbf{u}$ for the stacked thrust commands, execution is

$$\mathbf{u}^{\text{exec}} = \mathcal{F}_{\text{CBF}}(s, \mathbf{u}^{\text{nom}}). \quad (5)$$

The blending coefficient therefore characterizes nominal composition, not the final motion's proportional contribution from each source. Pursuit, lateral interception, and region maintenance are functions to identify in recorded trajectories. They are not roles assigned by large or small coefficients: Boids itself includes pursuit, and the learned branch receives task, formation, and collision feedback. Relaxed CBF feasibility is evaluated through measured collision outcomes.

C. Task reward

The defender reward retains the paper's task, formation, and collision terms. Target entry yields -100 to each defender; capture yields 100 to a defender within the capture radius, 50 within three capture radii, and zero task reward otherwise. Each colliding defender receives one -50 penalty. Writing $e_i = (p_A - p_i)/\|p_A - p_i\|$, $q = \sum_i e_i$, and $e_T = p_A/\|p_A\|$, the formation term is

$$r_i^{\text{form}} = \frac{1}{2} e_T^\top q / \max(\|q\|, \epsilon) - \|q\|/N. \quad (6)$$

Terms are evaluated on terminal transitions too, with the exact released boundary comparisons retained in code. The joint learner uses $\bar{r} = N^{-1} \sum_i r_i$. Return, defense success, and capture time are distinct quantities throughout the evaluation.

III. PREDICTIVE CONDITIONAL RESIDUAL COMPOSITION

A. Interception prior and pointwise residual

The predictive realization preserves a strong cooperative interception prior and selects residual capture maneuvers by their joint feedback consequences. Let $u_i^0(s)$ denote the command from a minimum-cost assignment to lateral interception goals and $u_i^p(s)$ a direct-pursuit command. The residual composition is

$$u_i(s; m) = u_i^0(s) + m_i(u_i^p(s) - u_i^0(s)), \quad (7)$$

$$m \in \{0, 1\}^N, \quad \|m\|_0 \leq 2.$$

The assignment is a control prior: assignment-based capture-time optimization already has a substantial precedent [27], as does combining feedback control with learned residuals [28]. Here the evaluated mechanism is the joint, sustained consequence of choosing which defenders depart from that prior. Six defenders give 22 compositions. Every 2 s, each composition is simulated with feedback for the 2-s commitment and a 20-s continuation under the same composition. The predictor ranks collision, breach, and estimated capture time lexicographically; nonzero residuals incur a fixed 0.15-s departure cost. Physical commands and the common CBF are recomputed every 0.2 s. The zero composition retains the prior exactly. Attacker agility is inferred causally from past measurements on the common range [1, 8]; the controller does not receive true agility or future disturbances.

The matched additive control uses the same candidates and computes the same 22 forecasts, but selects with

$$\tilde{C}(s, m) = C(s, 0) + \sum_i m_i [C(s, \mathbf{e}^{(i)}) - C(s, 0)], \quad (8)$$

where $\mathbf{e}^{(i)}$ is a singleton pursuit composition and C scalarizes predicted collision, breach, and time using weights 360, 180, and 1. This control removes joint-composition consequences from selection while preserving forecast budget. A short control retains joint selection but removes the 20-s continuation. The existing long predictor uses the same expanded agility-observer range; its original upper bound of 3.25 would be inappropriate for these conditions.

B. Conditional teammate response

Conditioning the remaining defenders' control roles on the selected capture maneuvers strengthens both task performance and the measured interaction effect. A pointwise pursuit residual leaves non-pursuing defenders in roles assigned under the assumption that the whole team is guarding. We instead complete the guard assignment after choosing the pursuing set $P(m) = \{i : m_i = 1\}$. Capture roles replace the $|P(m)|$ innermost lateral interception roles; remaining defenders are assigned to the remaining roles by minimum total arrival cost. Symmetric central-role ties are resolved by the smaller total assignment cost. Assignment itself is an established control prior [27]; the tested change is its conditioning on the residual composition and the joint evaluation of the resulting feedback.

Let $g^*(s, m)$ denote that assignment and $u_{i,g^*}(s)$ the corresponding guard command. The coordinated residual is

$$u_i^G(s, m) = \begin{cases} u_i^p(s), & m_i = 1, \\ u_{i,g^*}(s, m), & m_i = 0, \end{cases} \quad (9)$$

$$\delta u_i(s, m) = u_i^G(s, m) - u_i^0(s).$$

The zero composition exactly recovers the original joint interception prior. A one-bit composition change includes its induced teammate guard responses, so its value concerns a coordinated composition intervention. The same 22 binary compositions, 2-s commitment, 20-s predictive continuation, public observations, agility observer, and 0.2-s physical feedback are retained. Guard assignments are recomputed from the current state during both forecasts and execution. The matched additive comparator uses these same completed controls and all 22 forecasts, replacing joint-composition selection by the sum of single-defender cost changes defined above.

C. Adaptive response selection

To adapt the composition to the current state, we select the teammate response together with the pursuing set. Let $\rho \in \{0, 1\}$ indicate whether the remaining defenders reassign their guard roles. The resulting composition is

$$u_i(s; m, \rho) = u_i^0 + m_i(u_i^p - u_i^0) + \rho(1 - m_i)(u_i^G(s, m) - u_i^0). \quad (10)$$

The shared zero composition and 21 nonzero pursuit compositions in each response mode give 43 candidates. Joint prediction selects (m, ρ) under the same horizon and physical feedback. The matched additive control evaluates the same 43 candidates, reconstructs costs from single-defender differences within each response mode, and then selects across both modes. The matched short control also retains all 43 candidates.

IV. LEARNING COOPERATIVE CONTROL COMPOSITION

The learning realization below retains the original residual actor-critic formulation. Its local intervention fixes teammates' current coefficients, whereas the predictive operator above includes the induced guard response. Their outcomes are evaluated separately; predictive gains are not evidence of a learned-policy gain.

A. Candidate-control interaction-conditioned blending

The first stage independently generates learned candidates from the shared ARBoids observation encoder. Its task and Boids encoders and masked mean teammate embedding feed the retained 512-unit decision backbone. Reparameterized Gaussian samples are squashed by tanh to form ℓ_i . The second stage exchanges candidates from the same control instant. For each directed pair, a shared two-layer, 128-unit encoder receives

$$z_{ij} = (\delta p_{ij}^i/60, \sin \delta \psi_{ij}, \cos \delta \psi_{ij}, \delta v_{ij}^i/5, \delta r_{ij}, \ell_i, b_i, \ell_j, b_j), \quad (11)$$

where $b_i = (u_i^B - 250)/750$ and superscript i denotes rotation into defender i 's heading frame. For valid non-self neighbors $\mathcal{N}_i$, the candidate-interaction message and conditional blending policy are

$$h_i^{\text{rel}} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} f_{\kappa}(z_{ij}), \quad \theta_i \sim \pi_{\phi}^g(\cdot \mid h_i, \ell_i, b_i, h_i^{\text{rel}}). \quad (12)$$

An empty neighbor mask yields a zero message. This representation conditions composition on current candidate relations; it imposes no requirement that different defenders use different coefficients. The gate mean adds a message-dependent residual to the original adapter logit; the residual initializes at zero. A learned gate log standard deviation initializes at $\log(0.1)$.

With h_i the backbone feature and $\epsilon_i^g \sim \mathcal{N}(0, 1)$,

$$z_i^g = \mu_i^g(h_i, h_i^{\text{rel}}, \ell_i, b_i) + \sigma_i^g(h_i, h_i^{\text{rel}})\epsilon_i^g, \quad \theta_i = \frac{\tanh z_i^g + 1}{2}. \quad (13)$$

Thus the joint density factorizes into a proposal stage and a conditional gate stage,

$$\Pi_{\phi}(\ell, \Theta \mid o) = \prod_i \pi_{\phi}^L(\ell_i \mid o_i) \prod_i \pi_{\phi}^g(\theta_i \mid h_i, h_i^{\text{rel}}, \ell_i, b_i). \quad (14)$$

The gate log density includes both transformations:

$$\log \pi_i^g = \log \mathcal{N}(z_i^g; \mu_i^g, \sigma_i^g) - \log(1 - \tanh^2 z_i^g) + \log 2. \quad (15)$$

We evaluate the log-Jacobian in a stable softplus form. Gate losses backpropagate through the candidate messages when the proposal branch is unfrozen. Deterministic evaluation uses the transformed Gaussian means in both stages.

B. Joint configuration value and real-transition learning

For policy Π , let $Q^{\Pi}(s, \mathcal{A}, \Theta)$ denote the team's soft value conditioned on the current state, candidate controls, and complete composition configuration. Holding $\mathcal{A}$ fixed distinguishes a change in composition from an arbitrary replacement of the candidate controls. A local value can also be defined by averaging over teammates' coefficients under their policy. Retaining the full configuration enables the critic to represent differences across those teammate choices instead of only their average. This is value-level configuration interaction, distinct from the candidate-interaction message in (12).

In the implementation, u_i^L is determined by ℓ_i and the Boids command by s , so the same value is represented as $Q^{\Pi}(s, a)$ with $a = (\ell, \Theta)$. The transition includes the common CBF filter and subsequent vessel dynamics.

Each of two critics encodes a node containing a defender's centralized kinematic state, normalized composed nominal thrust, left-right thrust difference, and Boids candidate. The action coordinates are $v_i = \theta_i \ell_i + (1 - \theta_i) b_i$, where the corresponding physical thrust is $750v_i + 250$ N and b_i is the normalized Boids candidate. Equivalent proposal-gate factorizations therefore receive exactly the same value, as they produce the same input to the CBF; the current action's entropy is outside Q . A symmetric pair encoder takes the sum and absolute difference of node embeddings and pair distance. Masked means pool nodes and non-self pairs. The critic head

receives both pools, the intruder state, agility, remaining task time, and $N/7$, and outputs one team value. Shared encoders and symmetric aggregation give permutation invariance; the same networks accept two through seven defenders. Centralized information is restricted to critic training.

One replay record represents one complete team control cycle: observations, public kinematics, centralized state, proposals, all blending coefficients, actual filtered thrust, individual and mean rewards, next observations and states, and the terminal flag. Critic actions are (ℓ, Θ) ; filtered thrust is recorded for audit.

The entropy term uses the same per-defender scale as the mean team reward. For stage z , define

$$e_{\phi}(o, a; z) = \frac{\mathbb{I}[z = \text{joint}]}{N} \sum_{i=1}^N \log \pi_{\phi}^L(\ell_i \mid o_i). \quad (16)$$

During gate learning, the frozen proposal distribution supplies exogenous candidates and the gate optimizes physical task return. Joint learning regularizes the trainable proposal distribution at the per-defender scale. Gate sampling supports exploration, while its mean and variance learn through task value without a gate-density reward. For $a = (\ell, \Theta)$, our real-data evaluator and soft actor-critic [23] policy updates use

$$y = \bar{r} + \gamma(1 - d) \left[\frac{1}{2} \sum_{k=1,2} \bar{B}_k(s', a') - \alpha e_{\phi}(o', a'; z) \right], \quad (17)$$

$$L_B = \sum_k \mathbb{E}[(B_k(s, a) - \text{sg}(y))^2], \quad (18)$$

$$L_{\text{TD}} = \sum_k \mathbb{E}[(Q_k(s, a) - \text{sg}(y))^2], \quad (19)$$

$$L_{\pi} = \mathbb{E}[\alpha e_{\phi}(o, a; z) - \min_k Q_k(s, a)]. \quad (20)$$

The real-data evaluator averages its two heads for bootstrapping; policy improvement retains the main-critic minimum. Full, Same-info, No-candidate-interaction, Short, and Model-return share this choice. Truncated rollout tails use the same evaluator average, while the registered conditional-value diagnostics retain their original definition. We use $\gamma = 0.99$, target smoothing 0.005, and minibatches of 4096 joint transitions. Adam learning rates are 10^{-4} for critics and 10^{-5} for the actor and temperature. Proposal temperature starts at 0.2 and adapts during joint learning to a per-defender entropy target of -2 ; the gate stage has zero entropy cost and no temperature update. Each matched joint-learning arm has a main twin critic Q and an independent twin bootstrap evaluator B , both with the same 512/128 widths and initialization. After initialization from complete real-episode returns, the evaluator receives only L_B on real transitions, and its lagged copy follows $\bar{B} \leftarrow (1 - \tau)\bar{B} + \tau B$. The main critic receives the real-transition loss and the arm's prescribed auxiliary loss. Auxiliary gradients and main-critic parameters never update B or $\bar{B}$. Same-info retains the same evaluator updates; without auxiliary supervision its main and bootstrap critics remain identical. The additional evaluator updates are counted separately.

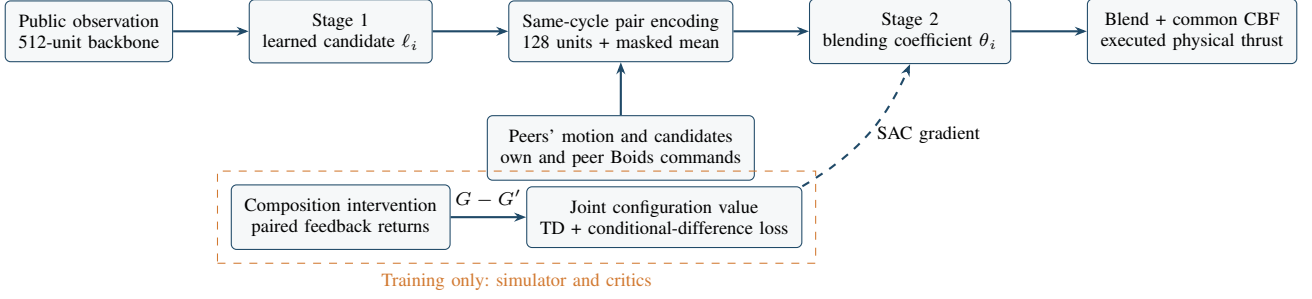


Fig. 2. The coupled policy and learning mechanism. Candidate-control relations condition local blending; paired composition interventions supervise conditional differences in the joint configuration value. Actor-critic updates connect that evaluation to the policy. Deployment uses the actor and common CBF.

C. Local composition intervention and conditional return difference

Let $\Theta^{(i)} = (\vartheta_i, \Theta_{-i})$ mean replacing component i by a reference coefficient ϑ_i while retaining the original indexing. The conditional value difference of this local composition change is

$$\Delta_i^\Pi(s, \mathcal{A}, \Theta; \vartheta_i) = Q^\Pi(s, \mathcal{A}, \Theta) - Q^\Pi(s, \mathcal{A}, \Theta^{(i)}). \quad (21)$$

For a captured training state, sample one common first joint action and a defender index i . Draw ϑ_i uniformly from $\{0, 0.5, 1\}$. The alternate first action $a^{(i)}$ fixes s , $\mathcal{A}$, and Θ_{-i} and changes only θ_i . Both branches start from separate complete copies of the same simulator state and share the same external random-number stream and policy sampling noise. From the second control cycle onward, every defender recomputes both policy stages using its own branch's evolved observations under the same frozen policy. The intervention is applied once; it does not fix the subsequent coefficients. The CBF, task reward, discount, and termination rules are identical in both branches.

Let H be the rollout horizon and $K \leq H$ the executed number of steps. Define

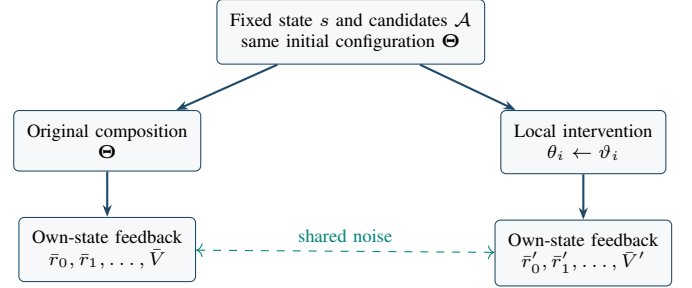
$$G_H(s, a) = \sum_{k=0}^{K-1} \gamma^k \bar{r}_k - \alpha \sum_{k=1}^{K-1} \gamma^k e_\phi(o_k, a_k; z) + \mathbb{I}[\text{not terminal at } K] \gamma^K \bar{V}(s_K), \quad (22)$$

$$\bar{V}(s_K) = \frac{1}{2} \sum_{j=1,2} \bar{B}_j(s_K, a_K) - \alpha e_\phi(o_K, a_K; z). \quad (23)$$

The prescribed first action has no entropy contribution, consistent with the soft Q convention. The subsequent action sample used for a truncated tail is shared by noise across branches. The actor, temperature, and real-data tail evaluator are frozen for an entire label batch, and the resulting targets stop gradients. Full, No-candidate-interaction, and Model-return use $H = 300$ and reach task termination, so their labels contain no learned tail; Short uses $H = 10$ and the truncated expression above. The sampled training target is

$$\widehat{\Delta G}_{H,i} = \frac{1}{R} \sum_{r=1}^R [G_H^{(r)}(s, a) - G_H^{(r)}(s, a^{(i)})]. \quad (24)$$

For training, $R = 8$: the state, first action, and intervention remain fixed across repetitions; future environment and policy



$$\Delta G_H = G_H(s, a) - G_H(s, a^{(i)})$$

Fig. 3. A local composition intervention fixes the current state, both candidate controls of every defender, and all other current coefficients. Later actions follow the same policy on each branch's own state. The label estimates a conditional soft-return difference for the composition variable.

noise are independent across repetitions and shared within each pair. The label combines closed-loop simulated rewards with a frozen value tail when the horizon is truncated. Its within-state Monte Carlo standard error is recorded. Thus the supervision contains simulated consequences of the composition change, rather than only a difference between the current critic's own predictions. The ordinary-return control uses the mean of each branch over the same eight repetitions.

D. Composition-specific conditional value supervision

The complete method adds

$$L_\Delta = \sum_{j=1}^2 \mathbb{E} \left[\left(Q_j(s, a) - Q_j(s, a^{(i)}) - \text{sg}\{\widehat{\Delta G}_{H,i}\} \right)^2 \right] \quad (25)$$

to the real-transition loss: $L_Q = L_{TD} + 0.1L_\Delta$. The ordinary model-return control uses

$$L_G = \frac{1}{2} \sum_j \mathbb{E} \left[(Q_j(s, a) - \text{sg } G_H(s, a))^2 + (Q_j(s, a^{(i)}) - \text{sg } G_H(s, a^{(i)}))^2 \right]. \quad (26)$$

Both objectives consume both branches. Each method generates labels under its own current frozen policy. The random-seed rule, snapshot sampling, intervention distribution, pair count, and horizon are matched; trajectories need not coincide after policies diverge. This avoids assigning another policy's

continuation return to a method’s own conditional value. Real transitions learn task value, paired consequences distinguish the conditional effect of local composition, and the actor objective transmits that evaluation to the blending policy. This supervision targets the composition variable and its closed-loop consequences within actor–critic learning; it does not introduce a separate role or responsibility allocator.

The long variants run to actual task termination, for at most $H = 300$ control cycles (the remaining portion of the 60-s episode); the short variant uses $H = 10$ (2 s). After critic warmup, every 1000 real transitions produces 64 fixed root interventions drawn from a rolling pool of 256 snapshots sampled every 16 transitions. Each root uses eight paired futures. The 64 averaged labels are reused between refreshes alongside the 4096-transition real minibatch, a ratio of 1/64; rollout generation does not cover all 4096 samples at every update. A new batch replaces the previous labels. No auxiliary rollout transition enters the ordinary replay. The implementation uses the complete training simulator for labels. Long-label error includes finite-sample variability; Short additionally depends on a bootstrapped tail. Real transitions, simulated transitions, gradient updates, and wall time are counted separately.

E. Training stages, recovery, and deployment

Before the first actor update at real step 5000, each matched joint-learning arm initializes its evaluator using completed episodes within the 4999 real transitions already collected under the fixed initial policy. Each transition records its reward and current entropy cost $c_t = \alpha e_\phi(o_t, a_t; \text{gate})$. Within a completed episode, targets obey $R_t = \bar{r}_t + \gamma(R_{t+1} - c_{t+1})$, with $R_T = \bar{r}_T$ on its terminal transition. This excludes the current action’s entropy and uses no learned value tail. The unfinished final episode is excluded. During these 5000 regression updates, the three control-input columns of each critic’s node encoder start and remain at zero: initialization fits a state-value prior. Both main and lagged critics are then initialized from the fitted evaluator; the main optimizer receives an independent copy of its state. All control-input columns are released for subsequent ordinary learning. Auxiliary labels and policy updates begin at step 5000. Every matched arm uses the same initializer. Counters, recorded entropy costs, and the private warmup sampling stream support exact recovery, and these updates are charged separately. Changes to this protocol require fresh matched training from the common pretrained actor.

Each seed first trains ARBoids with three defenders for one million real transitions. Every arm starts from the same seed-specific actor. Development uses at most 250000 additional transitions with proposal parameters frozen and agility uniform over 2.25 ± 0.5 ; candidate actions continue to depend on each current observation. The proposed formal budget is one million additional transitions in total: 250000 frozen-proposal transitions followed by 750000 joint transitions. Joint learning unfreezes the proposal parameters and activates their per-defender entropy target of -2 . The four joint-stage agility centers are 2.0, 2.25, 2.5, 2.75, each with half-width 0.5 for 187500 transitions. All five matched joint-learning arms share

TABLE I
FRESH ADAPTIVE-COMPOSITION CONFIRMATION, 256 PAIRED SCENARIOS

Controller	Time (s)	Captures	Successes
Adaptive joint	21.103	256	256
Adaptive additive	26.417	251	255
Adaptive short	30.245	244	253
Fixed initial roles	31.028	244	256
Existing long predictor	30.666	240	256
Fixed guard completion	19.814	256	256

this entropy objective and temperature schedule. ARBoids continuation retains its individual critic, gate exploration, and entropy implementation. Common pretraining and development costs are reported separately from formal training.

Resumable checkpoints include both critic pairs and their optimizers, the lagged evaluator, actor, temperature, phase, real replay, snapshot pool, auxiliary labels, environment, counters, and random states. A legacy development checkpoint with a self-coupled bootstrap cannot be silently resumed under the independent-evaluator protocol. Recovery holds its actor fixed, fits a fresh evaluator to existing real replay for a prescribed 10000 updates, and audits the resulting value tails against complete returns on 64 independent states with two continuations each. Only a passing calibration replaces the development critic and invalidates its cached labels; the actor, real replay, environment-step count, and training random state are retained. Recovery costs and historical training are development expenditure, not a new matched-from-initialization run. Same-info requires only an exact evaluator duplication. Redundant consecutive next states are stored by exact references with boundary exceptions; reconstruction preserves every float. A separate deployment artifact contains the actor and configuration only. VRX aligns vessel feedback to one control timestamp, generates candidates and gates, and applies the same CBF. The numerical left/right action ordering is mapped to Gazebo starboard/port thrust so yaw signs agree.

V. EXPERIMENTAL DESIGN AND RESULTS

A. Adaptive joint composition: held-out task evidence

The adaptive method and its four primary references were then frozen before 256 unused scenarios, again balanced between agility 4 and 6. Fixed completion was declared a secondary comparison for preservation of the preceding strong result, rather than another reference requiring a further 10% speed gain. Table I retains all six methods and all 1536 episodes. All six methods have zero collisions. Adaptive joint selection is 20.12%, 30.23%, 31.99%, and 31.18% faster than the four primary references. Accounting for the linked random streams in 128 blocks, the respective simultaneous 98.75% paired time intervals are $[-7.458, -3.248]$, $[-11.323, -7.019]$, $[-12.373, -7.532]$, and $[-11.905, -7.240]$ s. The dependence-aware binary bound uses that a positive two-stratum block difference is at least 1/2 and a negative one has magnitude at most one; two Clopper–Pearson tails of 0.025 bound the corresponding event probabilities. All capture and success lower bounds exceed -3 percent-

TABLE II
INDEPENDENT CONDITIONAL-GUARD EVALUATION, 128 PAIRED
SCENARIOS

Controller	Time (s)	Captures
Joint guard completion	19.858	128
Original joint predictor	24.888	126
Additive guard completion	25.027	127

age points; the smallest success bound is -2.841 points. Both the original registered analysis and this additional analysis, specified before inspecting the new task outcomes, pass all four combined criteria. The secondary fixed-completion controller is 1.289 s faster on this bank with identical observed outcomes. Thus the new bank establishes superiority over the primary references and retains a strong task result, but does not establish dominance over fixed completion.

The expanded bank admits an exact elapsed-time bound. Once a completed forecast with no predicted collision or breach supplies a cost B , an unfinished forecast cannot win if its elapsed time plus departure cost is strictly greater than B : its eventual nonfailure estimate is no smaller, and predicted failure is worse lexicographically. The original candidate order resolves ties. On two complete development trajectories, bounded parallel evaluation preserves all 189 physical commands and selected forecast trajectories elementwise. With other experiments stopped, 19 warm planning calls have median 94.13 ms, p95 191.61 ms, and maximum 217.87 ms, compared with unbounded parallel p95 346.23 ms. One bounded call exceeds the 200-ms interval. Ordinary feedback has p95 3.04 ms; cold startup and first prediction take 8.59 s. This is measured computation performance, not a hard real-time guarantee. Independent task comparisons use all complete forecasts for each method.

B. Conditional guard completion: strengthened interaction evidence

We evaluate the conditional teammate-response operator in Section III.

Following development on 16 scenarios, the method and analysis were frozen before 128 unused six-defender scenarios, balanced between attacker thrust multipliers 4 and 6. Initial states, disturbances, CBF filtering, and the 60-s capped capture metric are paired. All scenarios are retained. All three controllers achieve 128 defense successes and zero collisions. Joint guard completion reduces capped time by 20.21% relative to the original strong joint predictor and 20.65% relative to matched additive completion. The paired differences are -5.030 and -5.169 s, with simultaneous 97.5% stratified bootstrap intervals $[-7.411, -2.762]$ and $[-7.405, -2.998]$ s, using 20000 draws. Both agility strata improve: the new controller takes 18.412 and 21.303 s, versus 23.772 and 26.003 s for the original predictor and 23.397 and 26.656 s for additive completion. Both contrasts pass the predeclared combined gate of at least 10% mean time reduction, adjusted intervals below zero, and preserved task outcomes, including the one-sided 95%, -3 -percentage-point capture and success noninferiority

checks. This result establishes task and interaction gains for conditional guard completion; the earlier pointwise method's failed tests below retain their original conclusions.

A second independently frozen bank contains 256 further scenarios, again balanced between the two agility strata. Joint completion takes 19.863 s, versus 24.823 s for matched additive completion, 29.913 s for matched short prediction, 30.257 s for fixed initial roles, and 32.370 s for the existing long predictor. The respective improvements are 19.98%, 33.60%, 34.35%, and 38.64%; simultaneous 98.75% paired difference intervals are $[-6.845, -3.100]$, $[-12.545, -7.574]$, $[-12.683, -8.197]$, and $[-14.913, -10.071]$ s. Capture counts are 254, 254, 243, 246, and 230; defense successes are 254, 254, 255, 254, and 256, respectively. All methods have zero collisions. The time advantage and joint-versus-additive effect repeat, and all capture and success noninferiority checks pass. However, two target breaches make observed defense success lower than short and existing long prediction, so the full four-reference combined gate does not pass its additional observed-success requirement. The additive and fixed-role contrasts pass their complete criteria. These failures remain part of the reported evidence; diagnosing them does not convert the same scenarios into independent tests of a revised method.

As an inexpensive development challenge, directly assigning the nearest one or two defenders to pursue while completing the guard roles does not reproduce this benefit. With the same 2-s role commitment, their capped times are $33.150/32.825$ and $28.750/44.125$ s at agility 4/6, compared with $17.925/23.750$ s for joint predictive completion on those 16 development scenarios. The stronger-agility rule controls also lose captures and defense successes. These small development comparisons motivated the independent test; they are kept separate from its inferential results.

Reusing composition-dependent role layouts and computing each state-dependent candidate set once removes redundant work without changing the control law. All 16800 candidate commands checked across 720 physical states with four, six, or eight defenders agree elementwise with the frozen implementation. On two complete six-defender development trajectories, serial and 22-worker parallel execution also agree in all 167 physical commands and selected plans. With other experiments stopped, the 16 warm planning calls have parallel p95 latency 156.36 ms and maximum 160.28 ms, versus serial p95 2950.37 ms, on the same 20-core CPU allocation reported below. All warm calls meet the 200-ms control interval; ordinary feedback has p95 2.93 ms. Pool startup and first prediction take 8.52 s. These measurements establish timing feasibility on the measured trajectories, with cold startup reported separately.

An audit of the random streams identifies dependence across agility strata: a scenario's disturbance seed equals its initialization seed plus 10^6 , which is also the next stratum's same-index initialization seed. An additional conservative analysis therefore resamples same-index strata together. It preserves both first-bank time effects, with 97.5% intervals $[-7.117, -2.912]$ and $[-7.494, -2.925]$ s, and all four second-bank time intervals remain below zero. However, the first bank supplies only 64 independent stream blocks; its zero success difference

TABLE III
FROZEN PREDICTIVE-COMPOSITION EVALUATION: CAPPED TIME, 256
PAIRED SCENARIOS

Controller	Time (s)	Captures	Successes
Joint interception prior	31.146	244	253
Joint sustained prediction	25.281	256	256
Additive composition selection	27.331	250	255
Joint short prediction	29.997	245	254
Existing long predictor	31.312	237	256

has a conservative one-sided lower bound of -5.60 percentage points. In the second bank's 128 blocks, success lower bounds are approximately -5.4 to -5.5 percentage points. The original registered outputs above are retained, but their nominal noninferiority passes do not establish dependence-aware reliability within the prescribed three-point margin. The task-time and interaction benefits survive this audit; further confirmation uses disjoint initialization and disturbance seed ranges.

1) *Development of adaptive teammate response:* The two observed breaches motivate the adaptive response family in (10).

On the reused 256-scenario bank, adaptive joint selection takes 20.178 s, with 255 captures, 256 defense successes, and zero collisions; matched adaptive additive selection takes 25.435 s, with 252 captures, 255 successes, and zero collisions. Relative to fixed completion, mean time increases by 0.316 s while both observed breaches are removed. A late-expansion variant, which exposes the second response mode only after every guard forecast predicts failure, does not repair either breach and is discarded. These findings are development evidence.

C. Structured predictive compositions: completed task evidence

The pointwise reference in (7) retains the original guard assignment.

Before evaluation, the implementation, comparisons, and analysis were frozen. The independent bank contains 128 six-defender scenarios at agility 4 and 128 at agility 6, paired across all five methods. Agility is an attacker thrust multiplier, not speed in metres per second. Initial states, disturbances, 60-s task outcomes, and safety filtering are common. All scenarios are retained, and a failure to capture is charged 60 s. Every method has zero observed collisions. Joint sustained prediction reduces mean capped time by 18.83% relative to the interception prior, 15.72% relative to short prediction, and 19.26% relative to the existing long predictor. Their paired differences are -5.865 , -4.716 , and -6.030 s, with simultaneous 98.75% bootstrap intervals $[-8.309, -3.414]$, $[-6.852, -2.550]$, and $[-8.672, -3.377]$ s, respectively. The analysis resamples scenarios within agility strata with 20000 draws and controls the four time comparisons at familywise level 0.05.

The additive contrast improves by 7.50% (-2.050 s), but its adjusted interval is $[-4.106, 0.032]$ s. Consequently the pre-declared combined gate—at least 10% time reduction against

every reference, simultaneous intervals excluding zero, and preserved task outcomes—does not pass. All capture and success comparisons pass the separate one-sided 95% noninferiority criterion of -3 percentage points, using conservative paired binomial bounds.

A separately frozen interaction replication uses another 256 scenarios with the same two agility strata and unchanged controllers. Joint and additive selection take 25.972 and 26.445 s, with 252 and 253 captures; both achieve 256 defense successes and zero collisions. Their difference is -0.473 s (95% paired interval $[-2.263, 1.363]$ s), and the conservative capture-difference lower bound is -3.71 percentage points. This replication does not pass its prespecified interaction-benefit and capture-preservation criteria. It does not replace the earlier failed combined gate. A supplementary fixed-assignment controller on these same new scenarios takes 30.639 s, with 243 captures, 254 defense successes, and zero collisions. Joint prediction is 15.23% faster than this simple strong reference (-4.667 s; 95% paired interval $[-6.753, -2.613]$ s), with capture and success noninferiority retained. For the pointwise residual in (7), the task advantage over role assignment repeats, whereas its incremental joint-selection advantage over an additive predictor is not established.

These results belong to the structured online predictor. Learned conditional-value replacements tested on separate development scenes have not preserved its task advantage and do not establish a residual-RL benefit. A subsequent fitted-value pilot changes one composition coefficient for a 2-s block, then resumes the same frozen predictive policy in both branches. A 17-parameter correction fits the paired real-return difference relative to the model estimate; its matched control fits absolute returns from the same data. Each training state uses two independent paired futures. A development benefit of 0.85 s at correction strength 0.25 did not repeat: two independent training datasets evaluated on 128 shared new scenarios give conditional-versus-uncorrected and conditional-versus-absolute time differences of $+0.417$ and $+0.611$ s. Their crossed training-dataset/scenario bootstrap 97.5% intervals are $[-1.445, 2.167]$ and $[-1.052, 2.384]$ s. The analysis rule was recorded after launch but before reading these evaluation outcomes. Both training replicates and all scenarios are retained; this learned correction is not adopted. Increasing to eight paired futures on exactly the same training states also fails to improve control: the two conditional corrections take 27.150 and 27.503 s on the shared 64-scenario development bank, versus 25.472 s uncorrected. These additional development outcomes do not support further replication-only training for this linear correction. The released APF opponent is sensitive to defender ordering, which is held fixed and identical across methods; broader opponent robustness remains to be established.

Candidate forecasts can be executed concurrently without changing the decision rule. After other experiment processes stopped, a 22-worker implementation was compared with serial execution on four complete development trajectories: six and eight defenders at agility 4 and 6, on an Intel Xeon Platinum 8470Q host with a 20-core allocation. All 389 executed commands, selected compositions, and selected fore-

cast trajectories agree elementwise. Among the 22 warm six-defender planning calls, parallel p95 latency is 175.35 ms versus 2973.44 ms serially; the parallel maximum is 180.12 ms, below the 200-ms control interval. Its 194 ordinary feedback calls have p95 latency 2.87 ms. Eight-defender planning still has p95 latency 343.36 ms and 12 of 17 warm calls exceed 200 ms. Six-defender pool startup and first prediction take 8.73 s and are reported separately. This demonstrates preserved task behavior and timing feasibility on the measured six-defender trajectories, not a general deadline guarantee. No new VRX result is claimed.

D. Learning mechanism: fixed endpoints and matched comparisons

Development seeds are 42, 101; the separate formal seeds are 202, 303, 404, 505, 606. Existing development pretraining is reused only after protocol, configuration, seed, one-million-step endpoint, exit status, and provenance are verified. The development matrix contains Full, Same-info, No-candidate-interaction, Model-return, and Short. Formal comparisons add ARBoids+CBF continuation, giving six trained arms with matched additional real-step budgets. The entire method and comparison protocol must be frozen before this five-seed formal batch starts. Evaluation uses the prescribed one-million-step endpoint.

The complete method (Full) and Same-info share candidate information, the interaction encoder, blending-policy structure, joint critics, and the CBF. Same-info removes the conditional-return auxiliary objective and its rollout generation while retaining real-transition SAC. Their contrast tests the additional learning value of composition-specific supervision, with its simulation cost reported separately. The Model-return control matches the two-branch simulation protocol and budget but fits the absolute returns using L_G , isolating the difference objective from additional model exposure. Short uses the same difference objective with a shorter continuation. Same-info and the version without conditional-return supervision are one control, not two separately trained methods.

The Same-info versus ARBoids+CBF contrast evaluates the overall extension to interaction-conditioned joint learning. Candidate messages, relation encoding, critic structure, and learning details change together, so this contrast cannot assign its entire effect to the candidate encoder. No-candidate-interaction isolates the actor’s access to teammates’ candidates: mask only ℓ_j and b_j in (11), while retaining teammate state, own candidates, relative kinematics, encoder and policy widths, joint critics, conditional-return supervision, safety filtering, initialization, and training budget. This attribution comparison follows selection of a promising task-performing method.

Learning curves evaluate 100 independent validation episodes every 5000 real transitions. Numerical tests use agility 1.5, 2.0, 2.5, 3.0 with $N = 3$, and $N = 2, \dots, 7$ at agility 2.25. Each of these ten conditions uses 200 paired scenarios per training seed. Boids, the unfiltered pretrained ARBoids actor, and the existing adaptive long-horizon selector are supplementary references evaluated afresh under the same

60-s task protocol. The unfiltered and long-selector references use each seed’s common pretraining weights, without additional learning; Boids has no learned weights. The long-selector reference executes its existing serial candidate loop with compiled nominal dynamics, and its reported latency refers to that implementation. The earlier five-process timing result remains associated with the original 80-s study.

Exploration first seeks excellent cooperative-control performance from candidate interaction, strengthens promising methods, and then completes the evidence required for submission. Initial structural checks use 128 independently seeded states, split into 64 debug and 64 heldout states with balanced three- and six-defender strata. Zero intervention, branch exchange, focal-gate isolation, shared random streams, and snapshot integrity verify that the intended experiment is being executed. Subsequent exploration is directed by capped capture time, capture rate, success, and physical collisions. Complete ablations, cross-seed replication, high-precision value calibration, and mechanism proof follow promising task results. Previously completed work retains its actual budget and outcome; missing checkpoints are not reconstructed.

The current search compares four task-aligned gate-improvement candidates: single-vessel or joint-configuration interventions, each trained with three or six defenders. Their undiscounted reward is negative elapsed time, with unsuccessful early termination charged the remaining time to the 60-s deadline. Complete paired continuations under the deterministic deployment policy identify the better intervened gates; clipped absolute return gaps weight their regression targets, while unchanged gates are anchored to the generating configuration. This phase uses at most 64 actor updates per intervention batch and omits critic fitting because complete returns and direct gate improvement require no value tail or critic gradient. These are exploration alternatives, not a selected final method. Earlier long- and short-horizon checkpoints retain their observed regressions and receive no additional training budget. New fixed checkpoints at 10000, 25000, and 50000 transitions are screened on the same 32 scenarios per three- and six-defender condition. The shared pretrained ARBoids+CBF actor provides a screening reference; the final comparison requires matched additional training budgets. A separate fresh bank confirms a selected method. Learning stability and value diagnostics are used when they help improve performance or resolve an implementation concern. The eventual freeze binds architecture, objectives, horizon, supervision, curriculum, reward, observations, CBF, controls, budgets, seeds, endpoint rule, and final evaluation banks.

E. Configuration mechanism: conditional value, matching, and interaction

Mechanism tests use an independent set of 32 states per seed generated by the common pretrained actor. A single sampled root action is held fixed for each intervention test, and four paired full-environment continuations estimate its complete

TABLE IV
FIXED TRAINING AND AUXILIARY SETTINGS

Setting	Value
Common pretraining per seed	10^6 steps
Frozen-proposal stage per arm/seed	2.5×10^5 steps
Formal joint stage per arm/seed	7.5×10^5 steps
Backbone / relation width	512 / 128
Batch size / critic learning rate	4096 / 10^{-4}
Actor / temperature learning rate	10^{-5} / 10^{-5}
Discount / target smoothing	0.99 / 0.005
Auxiliary weight / pair count	0.1 / 64
Label refresh interval	1000 real steps
Paired futures per fixed intervention	8
Complete real-return initialization	5000 updates
Long / short horizon	To termination / 2 s

soft-return difference up to task termination. We report

$$\delta q_m = \min_j Q_j(s_m, a_m) - \min_j Q_j(s_m, a_m^{(i)}),$$

$$E_\Delta = \frac{1}{M} \sum_{m=1}^M \left| \delta q_m - \widehat{\Delta G}_m \right|. \quad (27)$$

Its interpretation is paired with actual capture and efficiency outcomes. Evaluation states and future-noise seeds are disjoint from training labels and validation scenarios.

Two implemented configuration diagnostics use a further independent pool of 32 states per formal seed, balanced across three and six defenders. At each state, freeze Full and form its candidate controls and original configuration Θ^C once. Hold the root candidates fixed, share disturbance and policy-noise streams across alternatives, and use four complete continuations per state. From the second control cycle onward, each alternative resumes that policy's own-state feedback. Coefficient replacements, vessel pairs, permutations, and the two trajectory examples per seed are chosen without reference to the outcomes. These diagnostics distinguish configuration matching from conditional interaction; they do not alter the training objective.

Configuration matching. Compare the original configuration with a uniform mean and a permutation of its coefficients:

$$\Theta^C, \quad \bar{\theta} \mathbf{1}, \quad P\Theta^C, \quad \bar{\theta} = \frac{1}{N} \sum_i \theta_i^C. \quad (28)$$

Only the coefficient values are reassigned by P ; vessel states, candidate controls, and identities remain fixed. The mean comparison tests the value of individualized composition. The permutation comparison preserves coefficient diversity and tests the value of matching coefficients to particular vessels and candidates. A loss under permutation can also arise from purely additive individual values, so it does not by itself establish non-additive interaction. This operation differs from relabeling all vessel data together, under which the critic is permutation invariant.

Conditional configuration interaction. For a defender pair (i, j) , use the original configuration and one replacement for each coefficient to construct four first-step configurations. With

s , $\mathcal{A}$, and $\Theta_{-\{i,j\}}$ suppressed, define the paired second-order finite difference

$$\widehat{\mathcal{L}}_{ij} = \widehat{G}(\theta_i, \theta_j) - \widehat{G}(\vartheta_i, \theta_j) - \widehat{G}(\theta_i, \vartheta_j) + \widehat{G}(\vartheta_i, \vartheta_j). \quad (29)$$

Here $\widehat{G}$ estimates the complete soft return under the common frozen continuation policy, using shared noise across all four branches and no truncated value tail. The contrast asks whether defender i 's composition-change value depends on defender j 's composition. For an additive conditional value $q_i(\theta_i) + q_j(\theta_j) + C$, its population counterpart is zero. Consistent differences relative to paired sampling uncertainty provide evidence of non-additive conditional interaction under the evaluated dynamics, safety filter, and feedback policy. They do not alone establish that the learned actor exploits that interaction; training controls and executed behavior address that question.

F. Behavior and task: composition to executed maneuvers

Endpoint reviews include a controlled candidate-response diagnostic on 32 separate validation states generated by the common pretrained actor. At each state, four draws resample one teammate's learned candidate from the evaluated checkpoint's proposal distribution. State, own candidates, Boids candidates, and gate noise remain fixed. We record the focal coefficient response and its nominal and filtered thrust effects. Filtering four command combinations separates the gate effect with peer commands fixed, the direct CBF response to the changed peer command with gates fixed, and their combined effect. This diagnostic establishes sensitivity; task benefit is assessed with the matched training control. It is implemented for the gate and joint reviews, independently of final test states.

The behavioral analysis follows the complete control chain,

$$(\mathcal{A}, \Theta) \longrightarrow \mathbf{u}^{\text{nom}} \longrightarrow \mathbf{u}^{\text{exec}} \longrightarrow \text{team trajectory}. \quad (30)$$

Jointly showing candidate disagreement, blending coefficients, nominal commands, filtered thrust, and trajectories makes the mechanism in (4) observable. Traces use common, pre-specified scenario identities and are interpreted alongside all task outcomes. Functional descriptions such as pursuit, lateral interception, or region maintenance follow the observed geometry and motion; coefficient thresholds do not assign these roles. The supplementary configuration diagnostics compare their induced trajectories and task outcomes as well as return differences. Their complete command-chain analysis remains part of the pending mechanism evaluation.

G. VRX and learned opponents

VRX [26] compares the complete method, ARBoids+CBF, and same-information SAC in open water and dock settings with three and six defenders. Each of four conditions has 100 episodes per method, with 20 contributed by each training seed. A valid collision or breakthrough is a task failure and remains in the results. A missing simulator bridge, invalid trajectory, or simulator error is an execution failure and blocks completion until resolved.

TABLE V
EVIDENCE GROUPS AND THE CLAIMS THEY ADDRESS

Evidence group	Comparisons or observations	Supported question
A: Overall task	Full versus ARBoids+CBF under common safety and comparable training budgets; capped capture time with CR, SR, and collision rate	Is the paired task benefit credible, repeatable, and meaningful in absolute seconds?
B: Conditional value	Full versus Same-info on independent complete-environment E_{env} ; linked task outcomes and state/seed sensitivity checks	Does conditional-consequence supervision improve value learning and connect to control benefit?
C: Candidate interaction	Full versus No-candidate-interaction in the prespecified six-defender condition; three defenders as the ordinary condition	Does explicit teammate-candidate information provide incremental benefit where interaction is stronger?
D: Composition mechanism	Original/permuted configurations, four-branch interventions, executed thrust, and fixed trajectory examples	Does configuration matching or conditional interaction have a direct, interpretable control consequence?

A separately gated robustness extension admits these three methods to five alternating defender/attacker phases of 500000 transitions each. The primary numerical, calibration, VRX, and runtime results are exported before this extension; its additional 37.5 million transitions require a stated unresolved robustness question and an explicit continuation decision. Defender phases are 1, 3, and 5; attacker phases are 2 and 4. Each opponent switch clears incompatible replay and rollout caches. Defender policies are evaluated against the APF intruder and the same frozen attackers obtained from the ARBoids+CBF branch for that seed. Learning during a defender phase uses a fixed opposing policy.

H. Learning-study metrics, uncertainty, and computation

The primary task metric is fixed as mean capped capture time,

$$Y_e = \begin{cases} t_e^{\text{capture}}, & \text{capture,} \\ 60 \text{ s,} & \text{otherwise,} \end{cases} \quad \bar{T}^{\text{cap}} = \frac{1}{N} \sum_{e=1}^N Y_e. \quad (31)$$

Every episode enters this denominator. The primary contrast is Full minus ARBoids+CBF in the three-defender, agility-2.25 condition; negative differences favor Full. Capture rate, defense success, and physical collision rate accompany the primary result. We also retain breach and timeout rates. The six-defender condition in the same arena at agility 2.25 is the prespecified strong-interaction test for Full versus No-candidate-interaction; an ordinary-condition difference need not be large. Both core conditions use 200 paired scenarios per method and training seed before broad task extensions.

All five independent formal training seeds (202, 303, 404, 505, 606) are retained at the fixed one-million-transition endpoint. We report each seed, method means and standard deviations, and paired effect sizes with 95% confidence intervals. Intervals use 10000 hierarchical bootstrap replicates: sample training seeds, then paired scenarios or states within each selected seed. Pairwise differences match scenario and seed identities. We do not require every seed to win, select favorable seeds, or impose a fixed percentage improvement. Task relevance is assessed from absolute seconds saved and joint capture, defense, and collision outcomes, with an explicit recorded interpretation; statistical significance alone is insufficient.

For conditional-value attribution, Full–Same-info E_{env} differences use a separate balanced set of 64 states per seed and four complete-environment continuations. Ten-percent trimmed paired-state effects and leave-one-training-seed-out effects expose dependence on unusual states or a single run. Credible value-error improvement must connect to learning efficiency, control selection, or task benefit; the implemented formal summary reports the paired task link. Same-info retains the same actor inputs, networks, joint critic and safety filter and removes only conditional-consequence supervision. No-candidate-interaction masks only the teammate learned and Boids candidates in the gate, retaining teammate state, joint critic, supervision, safety and training budget.

For configuration contrasts, four continuation repetitions are averaged within each state before resampling states within seeds. Permutation tests configuration matching; it does not itself establish nonadditivity. Four-branch contrasts test conditional interaction. We report both the signed mean and a cross-repeat estimate of the squared conditional mean, $[(\sum_r I_r)^2 - \sum_r I_r^2]/12$, to distinguish mixed-sign interactions from positive absolute-error noise. Criterion D requires beneficial configuration matching, credible nonzero four-branch interaction, and changed executed control together. Executed first-step thrust and two prespecified trajectory examples per seed connect these mechanisms to behavior. These analyses remain separate from task superiority. Curves aggregate seed means and do not treat evaluation episodes as independent training runs.

A promising method is selected through task performance and fresh three- and six-defender confirmation, then frozen before independent formal training. Systematic replication, attribution, conditional-value calibration, and mechanism tests establish whether its advantages are reproducible, why they arise, and where they apply. These full-scale evidence requirements do not precede exploratory method improvement. Recovered development runs retain their actual histories and cannot be relabeled as fresh formal replicates. A–D remain required for final learned-policy claims; the predictive evidence above does not substitute for them.

Common pretraining, additional real interactions, auxiliary simulation steps, and wall time are recorded separately. Inference latency is measured only after concurrent experiment

TABLE VI
EARLIER ONLINE-SELECTION EVIDENCE MOTIVATING COMPOSITION AND
LONG-TERM TASK EVALUATION; ORIGINAL 80-S PROTOCOL

Original method	Success	Capture	Collision	T_{80} (s)
Unfiltered ARBoids	146	146	878	70.248
ARBoids+CBF	1000	998	0	21.095
Matched short selector	993	992	0	20.681
Strong fixed 2-s selector	1008	1007	0	19.423
Adaptive long selector	1024	1022	0	17.087
Guard-oriented selector	1020	954	0	27.573
Fixed 1-s long selector	1024	1022	0	16.907

work ends, including the observation adapter, actor, and applicable safety filter. Network transport latency is outside this local measurement. Actual VRX control durations and deadline misses are also retained. Trajectory and gate figures use recorded executions of the corresponding policy; schematic figures are explicitly labeled.

I. Learning-study execution status

The learner, matched controls, configuration mean/permutation tests, four-branch return diagnostics, executed-control traces, and A–D evidence aggregation are implemented; formal results remain pending. The primary effect, all seed-level effects, state/seed sensitivity analyses, and mechanism contrasts are generated from fixed run artifacts. Integration checks verify execution and do not substitute for performance evidence.

J. Earlier evidence under its original protocol

Earlier online-selection experiments motivate both the research object and the learning signal. Under the same instantaneous risk objective, independent and joint coefficient selection achieved 403/512 and 433/512 successes, respectively, with collisions decreasing from 55 to 29. The comparison shows that organizing the same local control sources differently can change team outcomes, motivating cooperative control composition.

Table VI retains the preceding six-defender, 1024-scenario results under the 80-s source protocol. ARBoids+CBF, the matched short selector, and the adaptive long selector all recorded zero collisions, while their capped capture times were 21.095, 20.681, and 17.087 s. The long selector reduced capped time by 17.38% relative to the matched short selector. These results motivate evaluating sustained task consequences in addition to immediate safety. The long selector chose online among five whole-team feedback candidates using existing policies. The new experiments ask whether the proposed composition-specific learning mechanism can acquire useful choices within the policy; its results are reported separately under the 60-s protocol.

VI. DISCUSSION

The predictive results identify the importance of the composition operator itself. Conditional guard completion repeatedly strengthens the joint-versus-additive time advantage, while adaptive response selection retains a clear advantage over

the four primary references and repairs the two development breaches. The fixed-completion reference remains faster in the new bank, so these data do not establish that adaptation dominates completion in every task outcome. The measured execution-time reduction comes from exact forecast pruning and is distinct from the control mechanism’s task benefit.

The mechanism links two learning decisions: how to compose the currently available control candidates and how to evaluate a local composition change under teammates’ choices. Candidate-interaction messages support the first decision; paired closed-loop consequences supervise the second. The Same-info, Model-return, and Short controls distinguish conditional supervision from policy expression, extra model exposure, and continuation horizon. The supplementary actor ablation isolates peer-candidate information within the otherwise matched learner.

Configuration matching, conditional interaction, and learning benefit have different evidential roles. Coefficient permutation tests matching to vessels and candidates; four-branch finite differences examine non-additive conditional value; training controls test whether the proposed mechanism learns useful composition. The executed command chain then connects those decisions to actual maneuvers and task performance. Because the value targets include entropy and terminal rewards, their differences are not success probabilities or saved seconds. Calibration is assessed together with policy behavior and capture efficiency.

The environment state used to construct labels is privileged training information. Deployment requires present observations and synchronized candidate messages, with no access to the simulator snapshots. A common CBF controls the comparison’s safety layer, while relaxed feasibility and dynamics mismatch remain observable through collisions and task outcomes. VRX evaluates transfer within simulation. Claims about real vessel operations require separate physical experiments.

VII. CONCLUSION

Conditional teammate responses turn a local pursuit residual into a coordinated control composition. Joint closed-loop evaluation of this composition establishes task and interaction gains over matched additive and short-horizon controls. The adaptive realization captures all 256 intruders in its fresh six-defender bank, with zero breaches and collisions, and passes all four declared primary contrasts, including the dependence-aware sensitivity analysis. Its speed relative to fixed guard completion remains an explicit comparison. Residual actor-critic learning and paired conditional-return supervision are retained as a learning realization, but current learned corrections have not added stable task gains. The established numerical contribution is predictive conditional composition under the tested task conditions; learned-policy and physical-transfer claims require their own completed evidence.

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